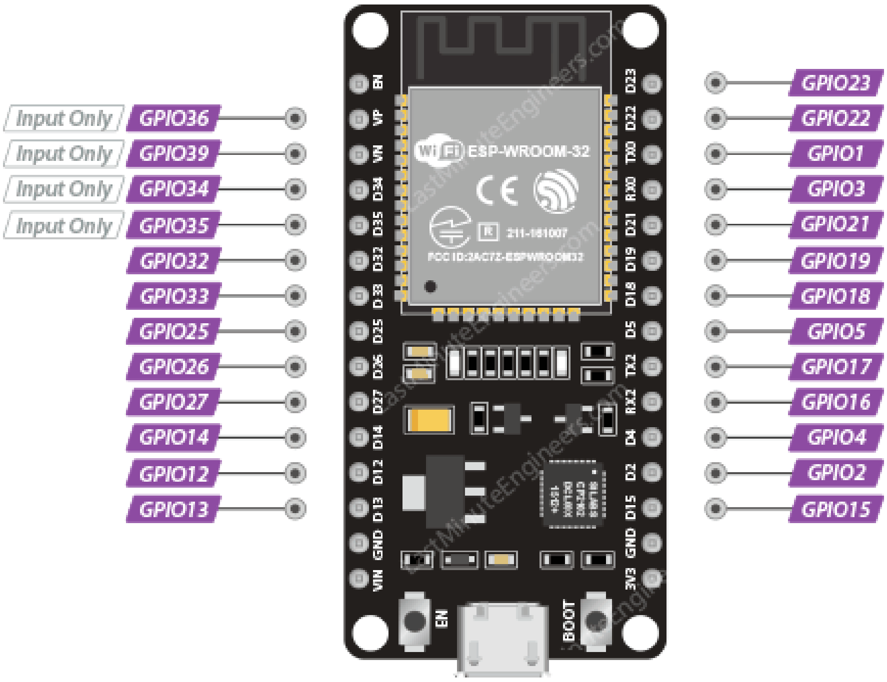
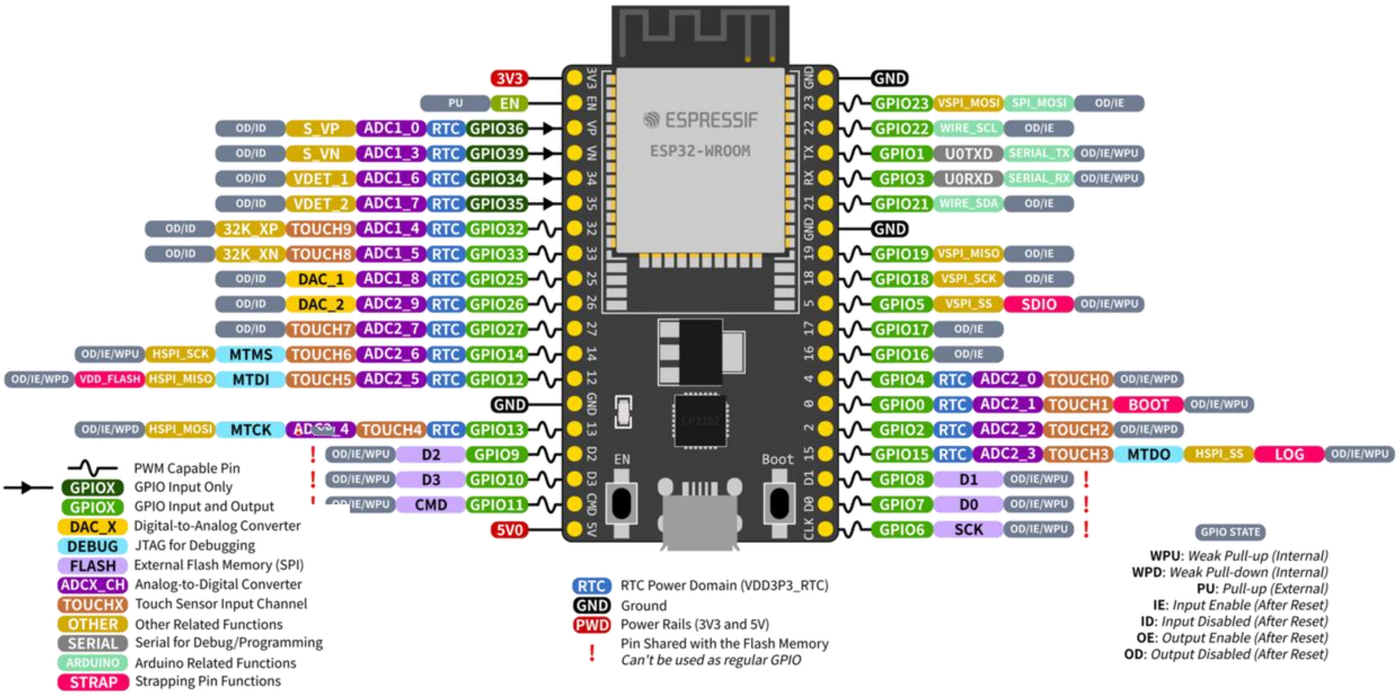
The [ESP32 DevKit V1]() and the [ESP32 Dev Kit C V4]() are both excellent choices for IoT projects. They share the same powerful dual-core processor, Wi-Fi, and Bluetooth capabilities, making them highly capable microcontrollers. However, they differ in their physical layout, pin availability, and manufacturer support.

Here is a detailed comparison of the two boards to help you decide:

ESP32 DevKit V1 Development Board:



ESP32 Dev Kit C V4:



Side by Side Comparison:

Board	**ESP32 DevKit V1 Development Board**	**ESP32 Dev Kit C V4**
Cost	**₹250–500**	**₹500–1200**
Manufacturer	Manufacturer Third-party (e.g., DOIT)	Manufacturer Espressif Systems (Official)
Pin Count	Pin Count 30 Pins	Pin Count 38 Pins
Module	Module ESP32-WROOM-32	Module ESP32-WROOM-32D / 32E
Built-in LED	Built-in LED Yes (GPIO 2)	Built-in LED Built-in LED availability depends on board revision.
Official Support	Official Support Community resources	Official Support Official Espressif documentation

Recommendation:

- **For Beginners:** The **ESP32 DevKit V1** is highly recommended if you want a budget-friendly option. It includes a **built-in programmable LED**, which is incredibly useful for testing code right out of the box without wiring external components.

- **For Advanced/Official Use:** The **ESP32 DevKit C V4** is the better choice if you want the **official Espressif design**. It features an updated module with **improved antenna performance** and a **38-pin layout** that exposes more GPIO pins for complex wiring.

Here's a practical conversion table between the two boards for the pins beginners use most often.

DevKit V1 Label	GPIO Number	DevKit C V4 Label
D0	GPIO0	0
D1	GPIO1 (TX0)	TX0 / 1
D2	GPIO2	2
D3	GPIO3 (RX0)	RX0 / 3
D4	GPIO4	4
D5	GPIO5	5
D12	GPIO12	12
D13	GPIO13	13
D14	GPIO14	14
D15	GPIO15	15
D18	GPIO18	18
D19	GPIO19	19
D21	GPIO21	21
D22	GPIO22	22
D23	GPIO23	23
D25	GPIO25	25
D26	GPIO26	26
D27	GPIO27	27
D32	GPIO32	32
D33	GPIO33	33
D34	GPIO34 (Input Only)	34
D35	GPIO35 (Input Only)	35
VP	GPIO36 (Input Only)	VP / 36
VN	GPIO39 (Input Only)	VN / 39

Common Labs

Component	GPIO	DevKit V1	DevKit C V4
DHT11/DHT22	GPIO4	D4	4
LED	GPIO2	D2	2
Buzzer	GPIO15	D15	15
Relay	GPIO23	D23	23
SDA (I2C)	GPIO21	D21	21
SCL (I2C)	GPIO22	D22	22
SPI MOSI	GPIO23	D23	23
SPI MISO	GPIO19	D19	19
SPI SCK	GPIO18	D18	18

Recommendation

Instead of saying:

```
D4
D2
D15
```

Say:

```
GPIO4
GPIO2
GPIO15
```

or

```
const int DHT_PIN = 4;
const int LED_PIN = 2;
```

Then you can use **either DevKit V1 or DevKit C V4** without changing code.

GPIOs Beginners Should Avoid Initially

GPIO	Reason
0	Boot pin
1	Serial TX
3	Serial RX
6-11	Connected to flash memory
34-39	Input only
12	Boot strapping pin (can cause boot issues)

- **Safe GPIOs for Labs**

2, 4, 5, 13, 14, 18, 19, 21, 22, 23, 25, 26, 27, 32, 33, GPIO16 and GPIO17 are also safe for most projects and are commonly used. *These are the pins I would use for almost all beginner ESP32 projects.*

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